Unit 5: The Pythagorean Theorem

Practice Problems — All Four Sections

Name: _____ Date: _____ Period: _____

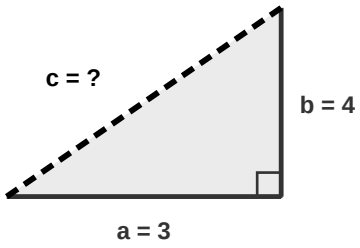
Key Formula: $a^2 + b^2 = c^2$ (a, b = legs; c = hypotenuse — always opposite the 90° angle)

5.1 — Introduction & How to Use

Determine if the missing side is a LEG or the HYPOTENUSE first, then solve.

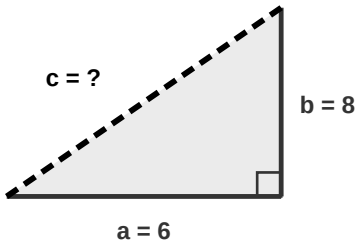
Part A — Solve for the Hypotenuse

Q1. A right triangle has legs 3 and 4. Find the hypotenuse.



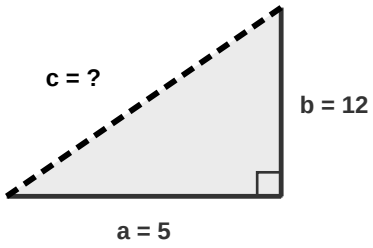
c =: _____

Q2. A right triangle has legs 6 and 8. Find the hypotenuse.



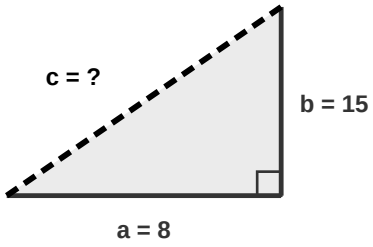
c =: _____

Q3. A right triangle has legs 5 and 12. Find the hypotenuse.



c =: _____

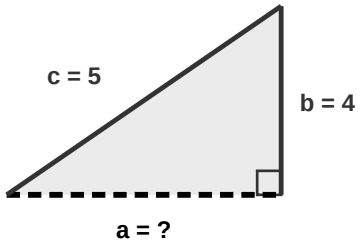
Q4. A right triangle has legs 8 and 15. Find the hypotenuse.



c =: _____

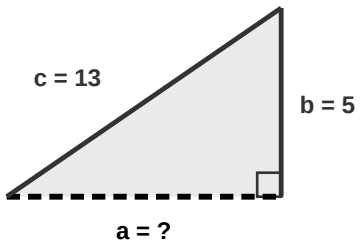
Part B — Solve for a Missing Leg

Q5. Hypotenuse = 5, leg = 4. Find the missing leg.



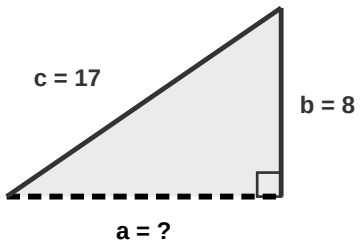
a =: _____

Q6. Hypotenuse = 13, leg = 5. Find the missing leg.



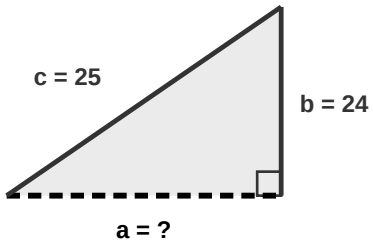
a =: _____

Q7. Hypotenuse = 17, leg = 8. Find the missing leg.



a =: _____

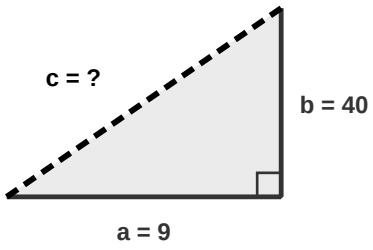
Q8. Hypotenuse = 25, leg = 24. Find the missing leg.



a =: _____

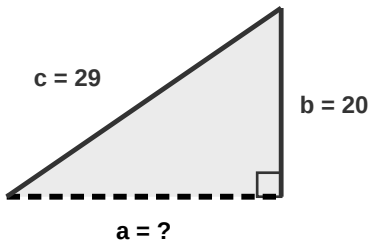
Part C — Identify the Missing Side, Then Solve

Q9. Legs are 9 and 40. Find the hypotenuse.



c =: _____

Q10. Hypotenuse = 29, leg = 20. Find the other leg.



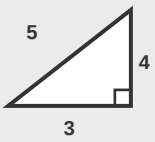
a =: _____

5.2 — Converse of the Pythagorean Theorem

Sort sides smallest to largest. Test: does $a^2 + b^2 = c^2$?

Part A — Classic Triples

Q1. Sides: 3, 4, 5. Is this a right triangle? Show your work.



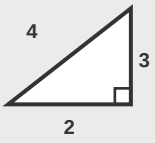
$3^2 + 4^2$ = ___

5^2 = ___

Equal? Yes / No

Yes / No: _____

Q2. Sides: 2, 3, 4. Is this a right triangle? Show your work.



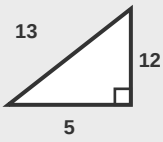
$2^2 + 3^2$ = ___

4^2 = ___

Equal? Yes / No

Yes / No: _____

Q3. Sides: 5, 12, 13. Is this a right triangle? Show your work.



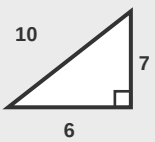
$5^2 + 12^2$ = ___

13^2 = ___

Equal? Yes / No

Yes / No: _____

Q4. Sides: 6, 7, 10. Is this a right triangle? Show your work.



$6^2 + 7^2$ = ___

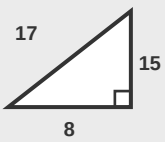
10^2 = ___

Equal? Yes / No

Yes / No: _____

Part B — Larger Numbers

Q5. Sides: 8, 15, 17. Is this a right triangle? Show your work.



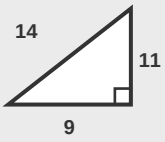
$8^2 + 15^2 =$ ___

$17^2 =$ ___

Equal? Yes / No

Yes / No: _____

Q6. Sides: 9, 11, 14. Is this a right triangle? Show your work.



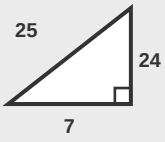
$9^2 + 11^2 =$ ___

$14^2 =$ ___

Equal? Yes / No

Yes / No: _____

Q7. Sides: 7, 24, 25. Is this a right triangle? Show your work.



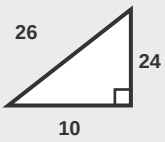
$7^2 + 24^2 =$ ___

$25^2 =$ ___

Equal? Yes / No

Yes / No: _____

Q8. Sides: 10, 24, 26. Is this a right triangle? Show your work.



$10^2 + 24^2 =$ ___

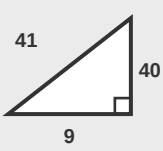
$26^2 =$ ___

Equal? Yes / No

Yes / No: _____

Part C — Tricky Values

Q9. Sides: 9, 40, 41. Is this a right triangle? Show your work.



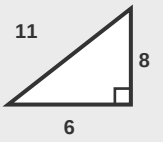
$9^2 + 40^2 =$ ___

$41^2 =$ ___

Equal? Yes / No

Yes / No: _____

Q10. Sides: 6, 8, 11. Is this a right triangle? Show your work. *(Careful — is this the same as 6-8-10?)*



$6^2 + 8^2 =$ ___

$11^2 =$ ___

Equal? Yes / No

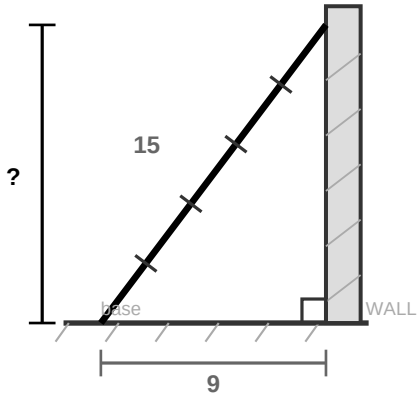
Yes / No: _____

5.3 — Word Problems

Label the diagram, identify the hypotenuse, then set up $a^2 + b^2 = c^2$ and solve.

Part A — Ladders, Ramps & Screens

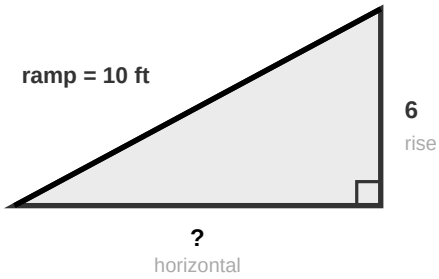
Q1. A ladder 15 ft long leans against a wall. The base is 9 ft from the wall. How high up the wall does it reach?



Height on wall =: _____

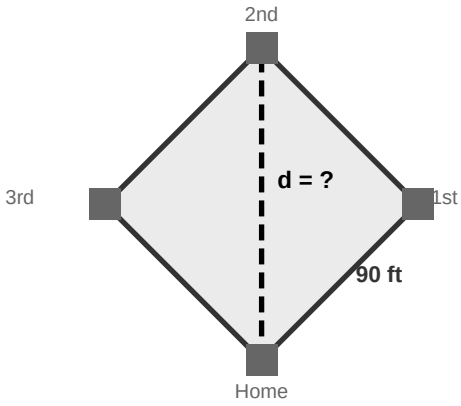
Wheelchair Ramp

Q2. A wheelchair ramp is 10 ft long and rises 6 ft vertically. What is the horizontal length?



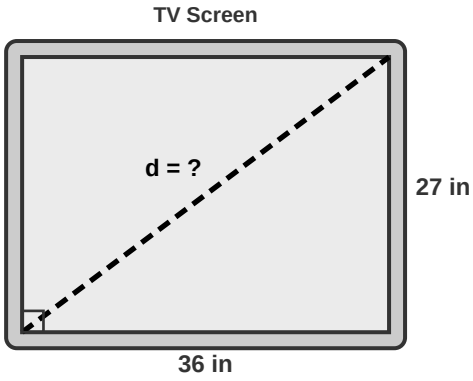
Horizontal length =: _____

Q3. A baseball diamond is a square with 90 ft sides. What is the distance from home plate to 2nd base? Round to the nearest foot.



Distance (nearest ft) =: _____

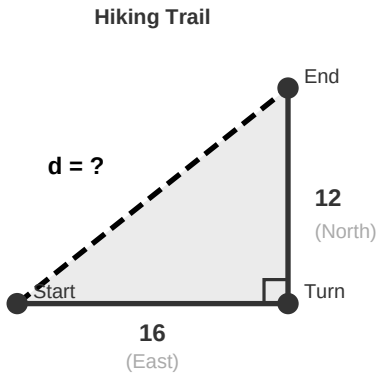
Q4. A TV screen is 36 in wide and 27 in tall. What is the diagonal measurement?



Diagonal =: _____

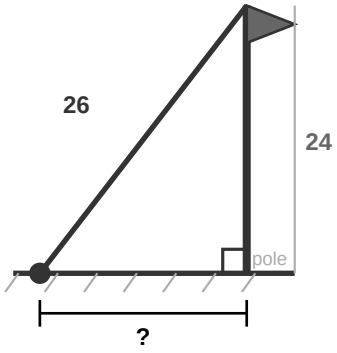
Part B — Navigation & Distance

Q5. A hiker walks 16 km East, then 12 km North. How far is she from the start in a straight line?



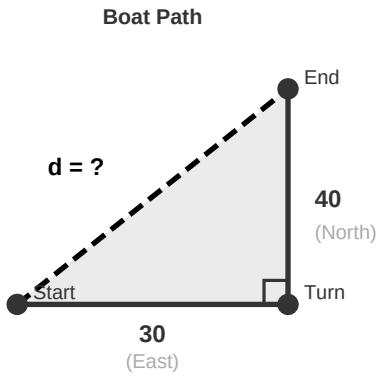
Straight-line distance =: _____

Q6. A flagpole is 24 ft tall. A wire runs from the top down to a stake in the ground. The wire is 26 ft long. How far is the stake from the base of the pole?



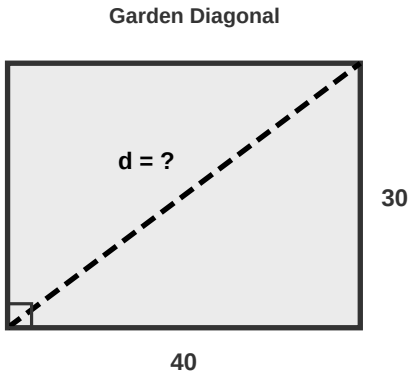
Distance from base =: _____

Q7. A boat travels 30 km East then 40 km North. How far is it from the dock in a straight line?



Straight-line distance =: _____

Q8. A rectangular garden is 40 m wide and 30 m long. How long is the diagonal path from one corner to the opposite corner?



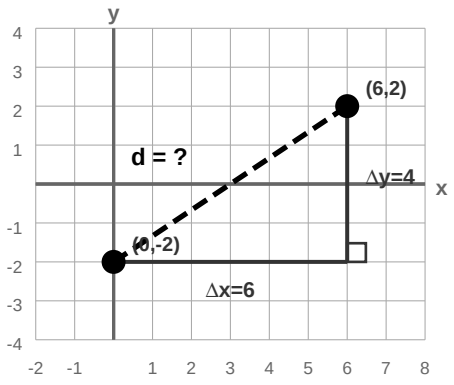
Diagonal =: _____

5.4 — Distance on the Coordinate Plane

Use the two points to form a right triangle, then apply the Pythagorean Theorem to find the distance.

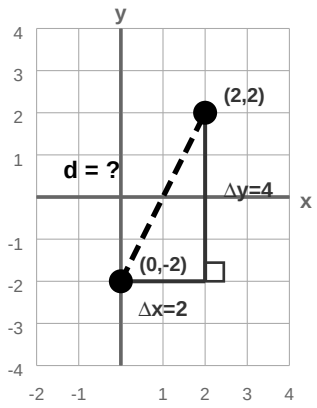
Part A — Integer Coordinates

Q1. Find the distance between (0, -2) and (6, 2). ∆x = ___ ∆y = ___



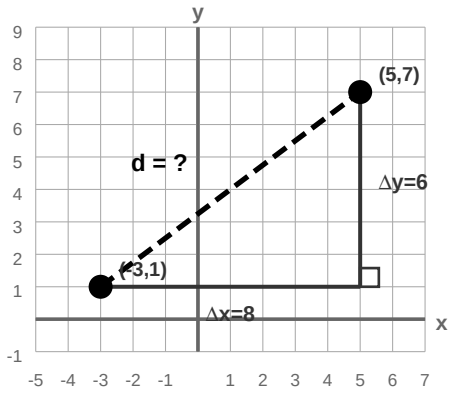
Distance: _____

Q2. Find the distance between (0, -2) and (2, 2). ∆x = ___ ∆y = ___



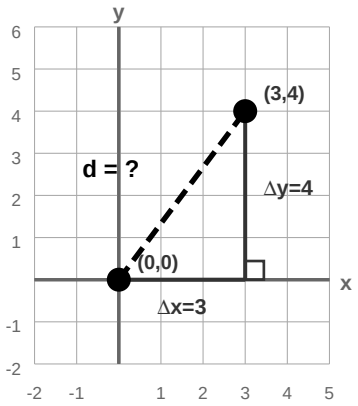
Distance: _____

Q3. Find the distance between (-3, 1) and (5, 7). ∆x = ___ ∆y = ___



Distance: _____

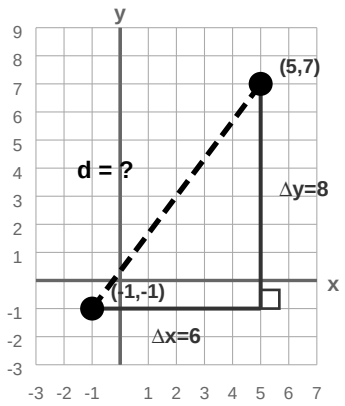
Q4. Find the distance between (0, 0) and (3, 4). ∆x = ___ ∆y = ___



Distance: _____

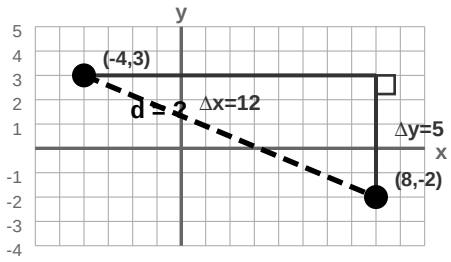
Part B — Negative Coordinates

Q5. Find the distance between (-1, -1) and (5, 7). ∆x = ___ ∆y = ___



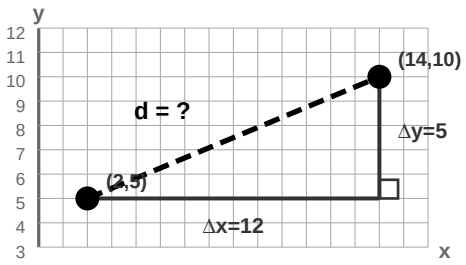
Distance: _____

Q6. Find the distance between (-4, 3) and (8, -2). ∆x = ___ ∆y = ___



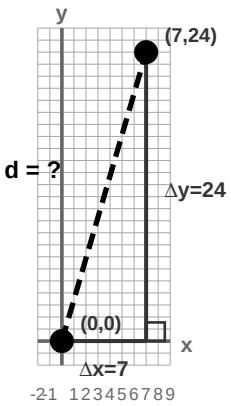
Distance: _____

Q7. Find the distance between (2, 5) and (14, 10). Each unit = 1 mile. ∆x = ___ ∆y = ___



Distance: _____

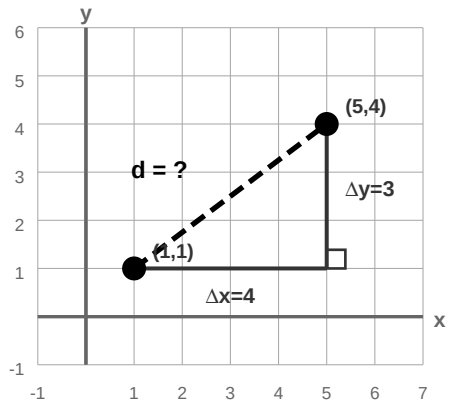
Q8. Find the distance between (0, 0) and (7, 24). ∆x = ___ ∆y = ___



Distance: _____

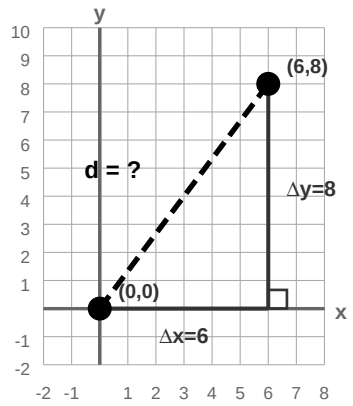
Part C — Applications

Q9. Find the distance between (1, 1) and (5, 4). Triangle A(1,1) B(5,1) C(5,4). Find AC. ∆x = ___ ∆y = ___



Distance: _____

Q10. Find the distance between (0, 0) and (6, 8). P(0,0), Q(6,0), R(6,8). Find PR. ∆x = ___ ∆y = ___



Distance: _____

ANSWER KEY

Unit 5: The Pythagorean Theorem — All Sections

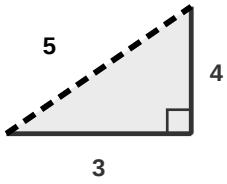
Section 5.1 — Introduction & How to Use

Q1. c = 5

$3^2+4^2 = 9+16 = 25$

$c = \sqrt{25} = 5$

→ *3-4-5 triple.*

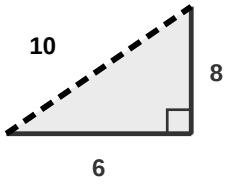


Q2. c = 10

$6^2+8^2 = 36+64 = 100$

$c = \sqrt{100} = 10$

→ *3-4-5 ×2.*

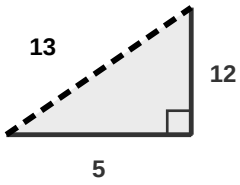


Q3. c = 13

$5^2+12^2 = 25+144 = 169$

$c = \sqrt{169} = 13$

→ *5-12-13 triple.*

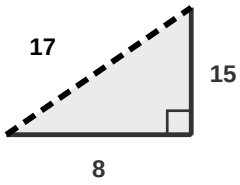


Q4. c = 17

$8^2+15^2 = 64+225 = 289$

$c = \sqrt{289} = 17$

→ *8-15-17 triple.*



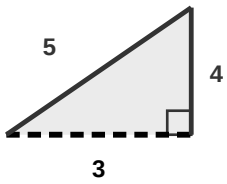
Q5. a = 3

$a^2+4^2 = 5^2$

$a^2 = 25-16 = 9$

$a = \sqrt{9} = 3$

→ *Subtract leg² from hyp².*



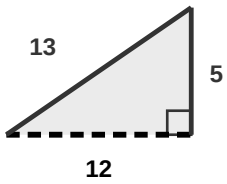
Q6. a = 12

$a^2+5^2 = 13^2$

$a^2 = 169-25 = 144$

$a = \sqrt{144} = 12$

→ *5-12-13 triple.*



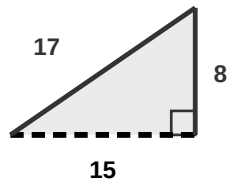
Q7. a = 15

$a^2+8^2 = 17^2$

$a^2 = 289-64 = 225$

$a = \sqrt{225} = 15$

→ 8-15-17 triple.



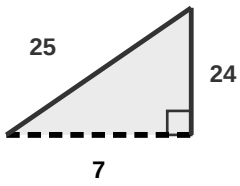
Q8. a = 7

$a^2+24^2 = 25^2$

$a^2 = 625-576 = 49$

$a = \sqrt{49} = 7$

→ 7-24-25 triple.

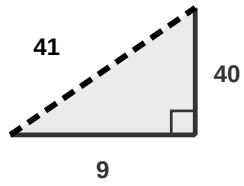


Q9. c = 41

$9^2+40^2 = 81+1600 = 1681$

$c = \sqrt{1681} = 41$

→ 9-40-41 triple.



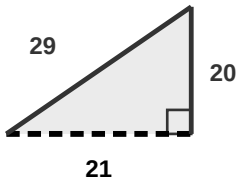
Q10. a = 21

$a^2+20^2 = 29^2$

$a^2 = 841-400 = 441$

$a = \sqrt{441} = 21$

→ 20-21-29 triple.

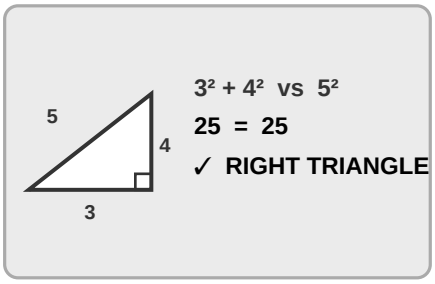


Section 5.2 — Converse

Q1. YES

$3^2+4^2=25$, $5^2=25$ ✓

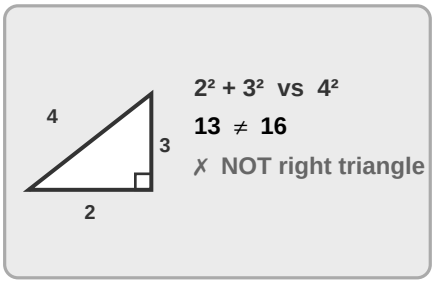
→ 3-4-5 triple.



Q2. NO

$2^2+3^2=13$, $4^2=16$ ✗ $(13\neq16)$

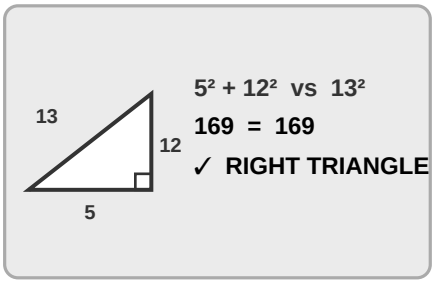
→ Not a triple.



Q3. YES

$5^2+12^2=169$, $13^2=169$ ✓

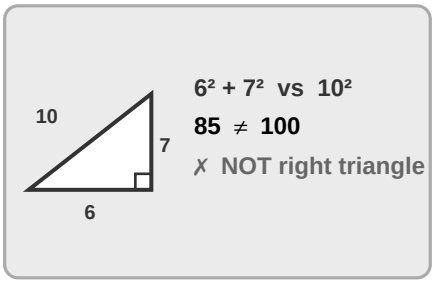
→ 5-12-13 triple.



Q4. NO

$6^2+7^2=85$, $10^2=100$ ✗

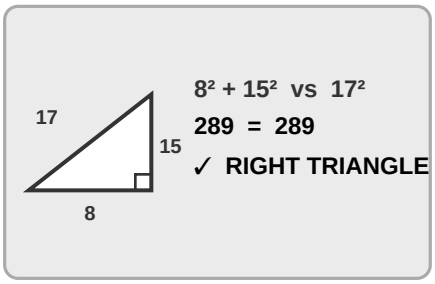
→ 85≠100.



Q5. YES

$8^2+15^2=289,\ 17^2=289$ ✓

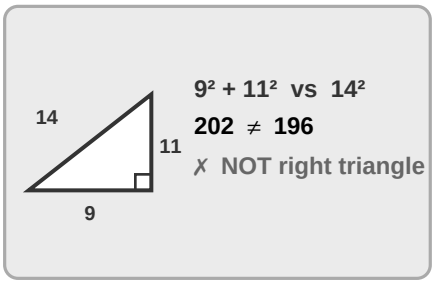
→ *8-15-17 triple.*



Q6. NO

$9^2+11^2=202,\ 14^2=196$ ✗

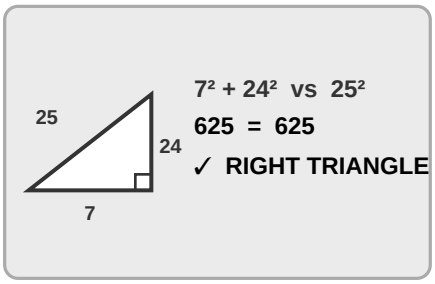
→ *202≠196.*



Q7. YES

$7^2+24^2=625,\ 25^2=625$ ✓

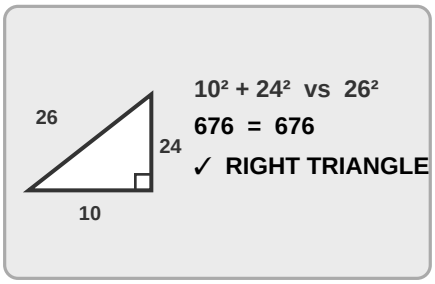
→ *7-24-25 triple.*



Q8. YES

$10^2+24^2=676,\ 26^2=676$ ✓

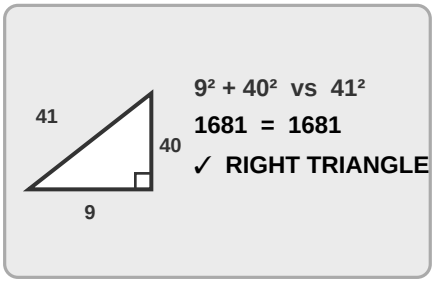
→ *5-12-13 ×2.*



Q9. YES

$9^2+40^2=1681,\ 41^2=1681$ ✓

→ *9-40-41 triple.*

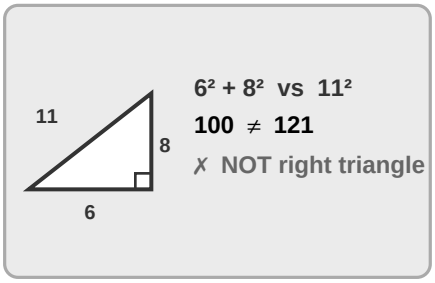


Q10. NO

$6^2+8^2=100,\ 11^2=121$ ✗

(6-8-10 works, not 6-8-11!)

→ *Trick!*



Section 5.3 — Word Problems

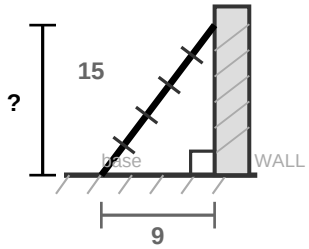
Q1. h = 12 ft

$9^2+h^2=15^2$

$81+h^2=225$

$h^2=144 \rightarrow h=12$

→ *3-4-5×3. Ladder=hyp.*

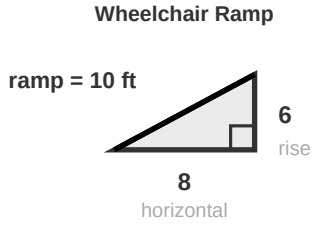


Q2. 8 ft

$a^2+6^2=10^2$

$a^2=64 \rightarrow a=8$

→ *6-8-10 triple.*

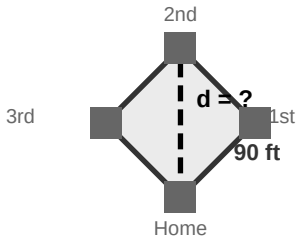


Q3. ≈ 127 ft

$d^2=90^2+90^2=16200$

$d=90\sqrt{2}\approx 90\times 1.414\approx 127.3$

→ *Square diagonal = s√2.*

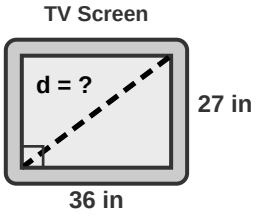


Q4. 45 in

$36^2+27^2=1296+729=2025$

$d=\sqrt{2025}=45$

→ *3-4-5×9.*

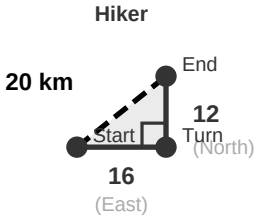


Q5. 20 km

$16^2+12^2=256+144=400$

$d=\sqrt{400}=20$

→ *3-4-5×4.*

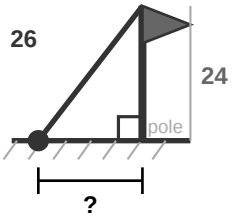


Q6. 10 ft

$a^2+24^2=26^2$

$a^2=676-576=100 \rightarrow a=10$

→ *5-12-13×2.*

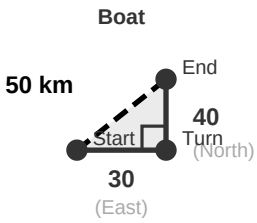


Q7. 50 km

$30^2+40^2=900+1600=2500$

$d=\sqrt{2500}=50$

→ *3-4-5×10.*

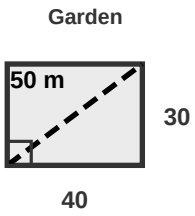


Q8. 50 m

$40^2+30^2=1600+900=2500$

$d=\sqrt{2500}=50$

→ *3-4-5×10.*



Section 5.4 — Coordinate Plane Distance

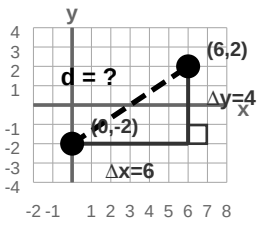
Q1. d=√52≈7.21

$\Delta x=6,\ \Delta y=4$

$d^2=36+16=52$

$d=\sqrt{52}=2\sqrt{13}\approx 7.21$

→ *Irrational — no whole number answer.*



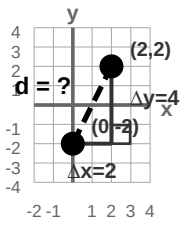
Q2. d=√20≈4.47

$\Delta x=2,\ \Delta y=4$

$d^2=4+16=20$

$d=\sqrt{20}=2\sqrt{5}\approx 4.47$

→ *Irrational.*



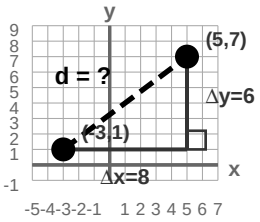
Q3. d = 10

$\Delta x=8,\ \Delta y=6$

$d^2=64+36=100$

$d=10$

→ *6-8-10 triple!*



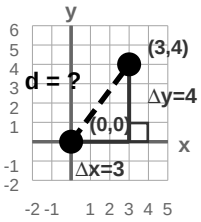
Q4. d = 5

$\Delta x=3,\ \Delta y=4$

$d^2=9+16=25$

$d=5$

→ *3-4-5 triple.*



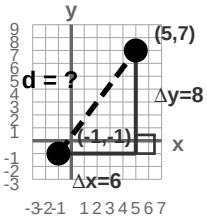
Q5. d = 10

$\Delta x=5-(-1)=6,\ \Delta y=7-(-1)=8$

$d^2=36+64=100$

$d=10$

→ *Subtract negatives carefully. 6-8-10 triple.*



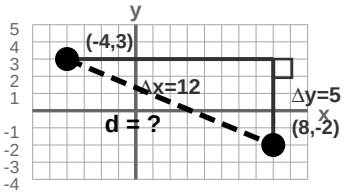
Q6. d = 13

$\Delta x=8-(-4)=12,\ \Delta y=|-2-3|=5$

$d^2=144+25=169$

$d=13$

→ *5-12-13 triple.*



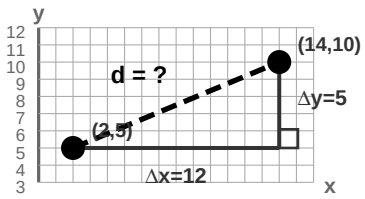
Q7. 13 miles

$\Delta x=12,\ \Delta y=5$

$d^2=144+25=169$

$d=13$ miles

→ 5-12-13 triple.



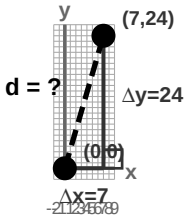
Q8. d = 25

$\Delta x=7,\ \Delta y=24$

$d^2=49+576=625$

$d=25$

→ 7-24-25 triple.

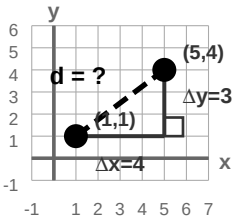


Q9. AC = 5

AB=4 (horiz), BC=3 (vert)

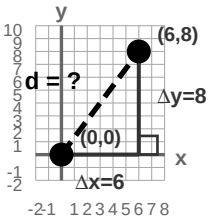
$AC^2=16+9=25 \rightarrow AC=5$

→ 3-4-5 triple.



Q10. PR = 10

PQ=6, QR=8

$PR^2=36+64=100 \rightarrow PR=10$

→ Right angle at Q(6,0).